CartCoders

Project Nova

BITS Pilani, Pilani Campus

Made by CartCoders

Problem Statement

Many gig economy workers, including Grab's most reliable driver and merchant-partners, are "credit invisible." They have consistent income streams but often lack the formal credit histories (e.g., from traditional bank loans or credit cards) that are required to access standard financial products. This creates a significant barrier, preventing them from securing loans to invest in their business—like upgrading a vehicle or expanding a food stall—or managing personal financial emergencies, thereby limiting their economic mobility.

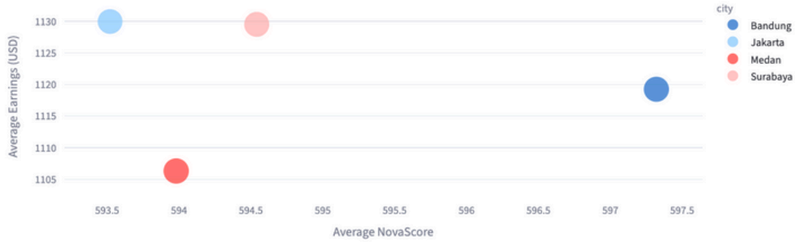
Market Insights

Assumption

We generated a 10,000-record dataset of 1.7 MB with the use of python library Faker which helped us generate a realistic database of Grab Partners. Here is a market analysis of Indonesia, which is a home to 51% Grab Partners

🏙️ Geographic Distribution

Cities: Score vs Earnings (Size = Partner Count)



City Performance Breakdown:

	city	Partners	Avg_Score	Avg_Earnings
0	Bandung	1,285	597	$1119
1	Jakarta	1,240	594	$1130
2	Medan	1,305	594	$1106
3	Surabaya	1,264	595	$1130

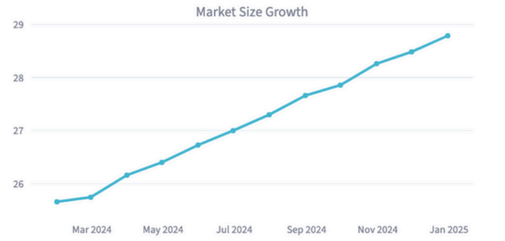
🚗 Vehicle Type Analysis

Vehicle Type Distribution

Performance by Vehicle Type

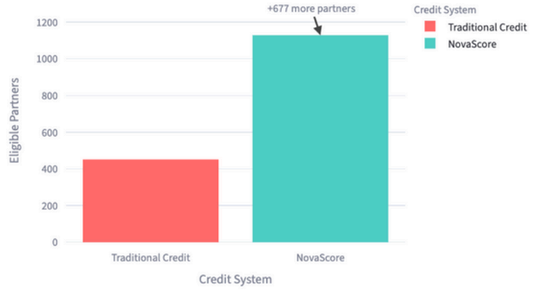
📈 Market Growth Analysis

12-Month Growth Projections

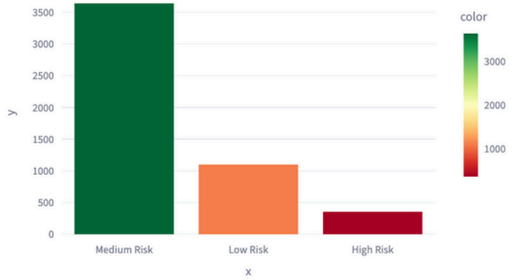


💳 Credit Accessibility Impact

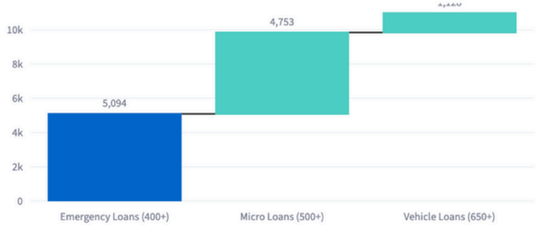
Loan Eligibility Comparison

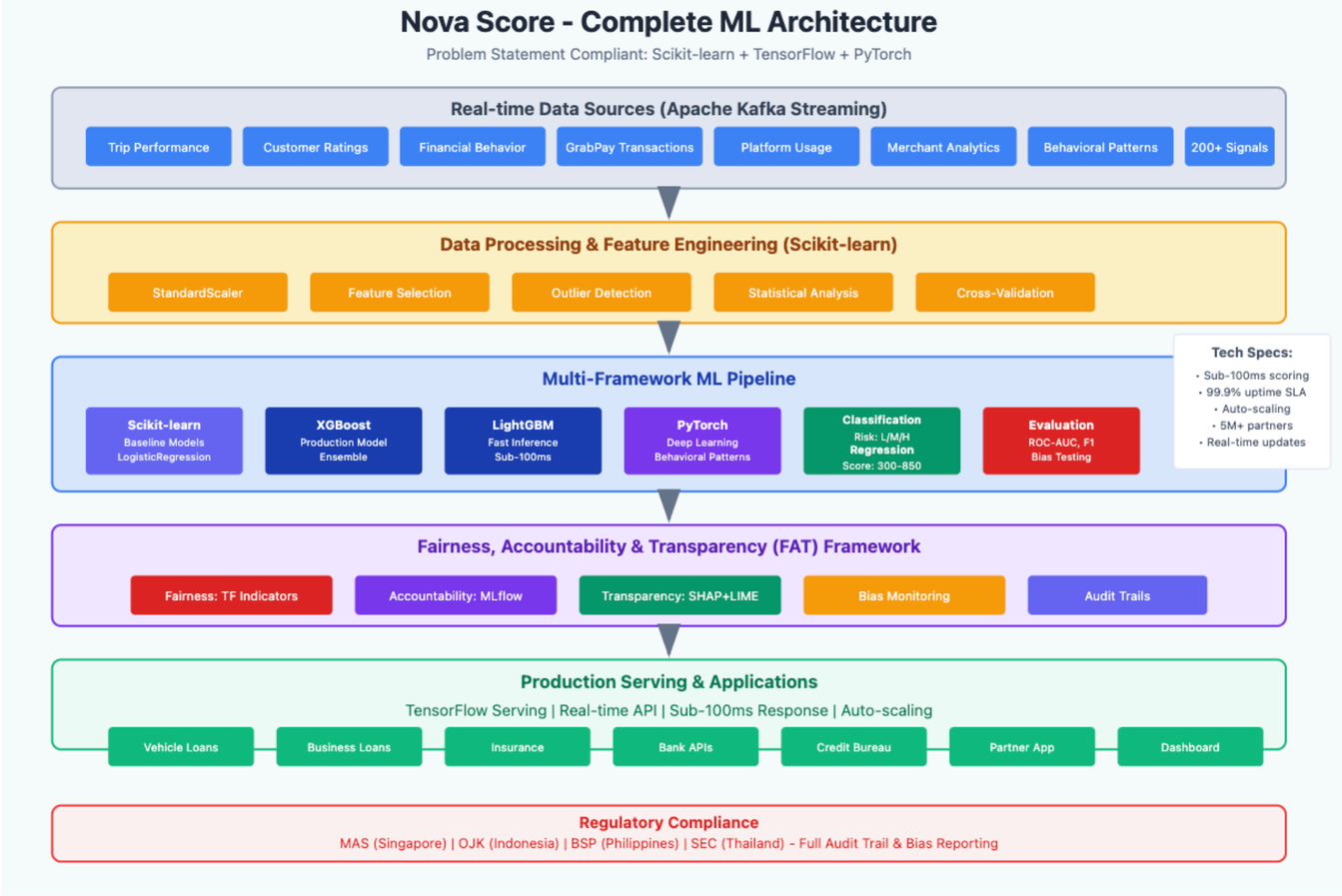


Risk Category Distribution



Loan Product Accessibility





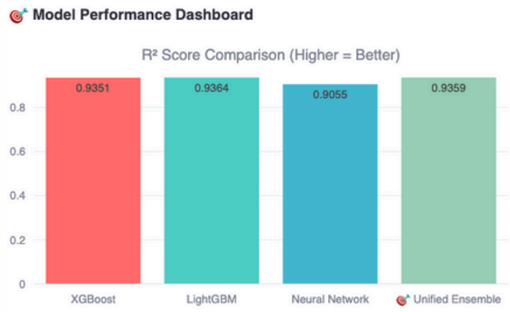
ML Architecture

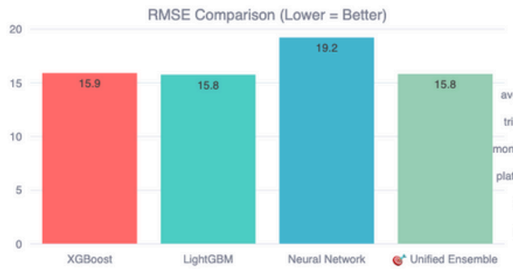
- Here's our 26 featured model comparison: logistic regression gave us a baseline, but our dual-output XGBoost–LightGBM ensemble — regression for NovaScore and classification for Risk Factor — hit 93.6% accuracy with an R^2 of about 93%, and response times under 100 ms using a Flask API.

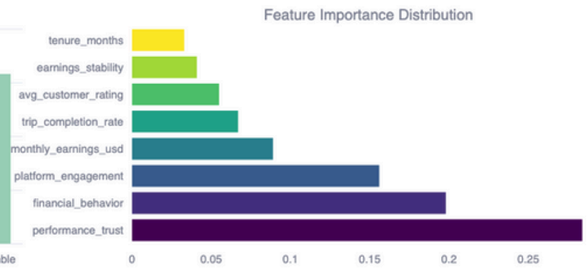
- We trained a dynamic scoring model that evolves with every ride, delivery, and rating to deliver 90% accurate predictions in real time. Also we used SHAP and LIME so every NovaScore is not just given, but explained.

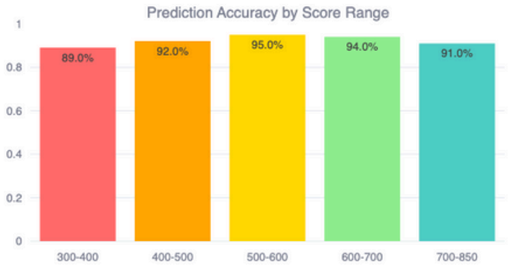
Model Performance

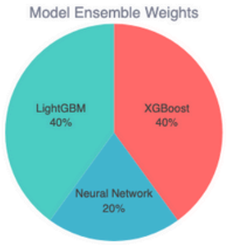
Solution

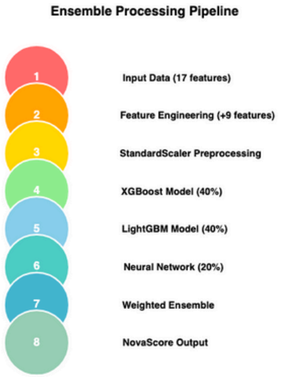












📊 Detailed Performance Metrics

	Model	R^2 Score	RMSE	MAE	Inference Time	Status	Cross-Val Mean	Cross-Val Std
0	XGBoost	0.9351	15.92	11.8	4ms	Production ✅	0.935	0.002
1	LightGBM	0.9364	15.76	11.5	2ms	Fastest ⚡	0.937	0.003
2	Neural Network	0.9055	19.24	14.2	12ms	Neural Power 🧠	0.906	0.004
3	🎯 Unified Ensemble	0.9359	15.84	11.7	6ms	Best Performance 🏆	0.936	0.002

R^2 of about 93% RSME of about 15.84 MAE of about 11.7

Live Predictions

Implementation

We use a 3-Pillar Strategy which weighs Performance Trust (40%), Financial Behaviour (35%), Platform Engagement (25%) to calculate NovaScore

🚗 **Trip Performance**

Trip Completion Rate: 0.92

Average Customer Rating: 4.60

Consistency Score: 0.85

Cancellation Rate: 0.04

💰 **Financial Metrics**

Monthly Earnings (USD): 800

Earning Volatility: 0.15

Total Trips: 800

Last 30 Days Trips: 65

📱 **Platform Engagement**

Tenure (Months): 15

Feature Adoption Score: 0.80

Active Days/Month: 25

GrabPay Usage: 0.70

Peak Hours Utilization: 0.60

Weekend Activity: 0.50

Referral Count: 8

📊 **Recent Trends**

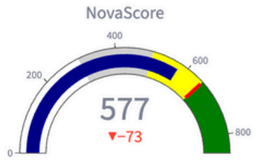
Recent Rating Trend: 0.08

Recent Earnings Trend: 0.06

🚀 Generate NovaScore Prediction

🎯 NovaScore

577



⚖️ Risk Category

🟡 Medium

⚠️ Medium Risk - Good creditworthiness

💳 **Loan Eligibility:**

✅ Vehicle Loans Available

Max Amount: $25,000

Interest Rate: 8.5%

Term: 48 months

🔍 **Prediction Insights**

Score Analysis:

- Your score of 577 is in the 50th percentile
- Risk Category: Medium
- Primary factors: Performance (92.0%), Earnings ($800), Experience (15 months)

Improvement Opportunities: - Need 73 more points for premium loans ($50,000 at 5.5%)

- Improve trip completion rate (currently 92.0%)
- Improve customer ratings (currently 4.6/5.0)
- Reduce cancellation rate (currently 4.0%)
- Increase platform feature adoption
- Focus on increasing monthly earnings

🧠 Individual Model Predictions

🌲 XGBoost: 566

🚀 LightGBM: 560

🧠 Neural Network: 608

Ensemble weights: XGB(0.4) + LGB(0.4) + NN(0.2)

🚨 **Emergency Loans (300-399)**

- Just getting started
- Building basic track record
- Max: $2,000 at 18%
- Expected: 350-390

💰 **Micro Loans (400-499)**

- Developing consistency
- Basic platform engagement
- Max: $10,000 at 12%
- Expected: 420-480

🚗 **Vehicle Loans (500-649)**

- Solid performance
- Regular earnings
- Max: $25,000 at 8.5%
- Expected: 550-620

🏆 **Premium Loans (650+)**

- Exceptional performance
- Max: $50,000 at 5.5%
- Extended terms (60+ months)
- Expected: 650-720

Fairness & Bias

Other Deliverables

We focus on mitigating potential biases sources like Digital Divide, Geographic Disparity, Gender Equity, etc..

Bias Detection & Mitigation

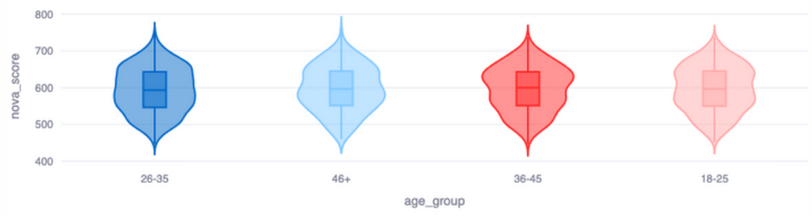
Potential Bias Sources

Bias Risk Assessment by Stage

Mitigation Techniques

Bias Mitigation Effectiveness

NovaScore Distribution by Age Group



Age Group Statistics:

age_group	mean	std
18-25	596.1	61.9
26-35	593.1	61.6
36-45	595.9	60.4
46+	595.8	60.8

NovaScore Distribution by Gender

Regulatory Compliance

Singapore (MAS) Indonesia (OJK) Philippines (BSP) Thailand (SEC)

Singapore (MAS) Compliance Status

Model Validation: ✅ Compliant

Bias Testing: ✅ Compliant

Explainability: ✅ Compliant

Data Protection: ✅ Compliant

Singapore (MAS) Indonesia (OJK) Philippines (BSP) Thailand (SEC)

Indonesia (OJK) Compliance Status

Credit Scoring Approval: ✅ Compliant

Consumer Protection: ✅ Compliant

Data Localization: ✅ Compliant

Regular Review: ✅ Compliant

Singapore (MAS) Indonesia (OJK) Philippines (BSP) Thailand (SEC)

Philippines (BSP) Compliance Status

Credit Risk Management: ✅ Compliant

Fair Lending: ✅ Compliant

Data Privacy: ✅ Compliant

Documentation: ✅ Compliant

Singapore (MAS) Indonesia (OJK) Philippines (BSP) Thailand (SEC)

Thailand (SEC) Compliance Status

Alternative Credit Approval: ✅ Compliant

Consumer Disclosure: ✅ Compliant

Model Risk Management: ✅ Compliant

Audit Trail: ✅ Compliant

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Try it out yourself

We built NovaScore, a proof of concept that turns partner trust and behaviour into a dynamic credit score. This area of focus is more important because for the first time, credit scoring isn't static — our model makes it dynamic, explainable, and fair, so every Grab partner's effort directly improves their financial future. We trained a dynamic scoring model that evolves with every ride, delivery, and rating to deliver 90% accurate predictions in real time. Also we used SHAP and LIME so every NovaScore is not just given, but explained.

Download our Dashboard, API & Dataset from here :

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Meet Our Team

Moksh Gupta

B.E. Computer Science

(Passing 2026)

Anusha Kansal

B.E. Electronics & Instrumentation

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Thank You